

● BOB AI HACKATHON SUBMISSION • TRACK: AI

D1 Mission Readiness & Predictive Maintenance Copilot

Condition-Based Maintenance (CBM+) powered by GPU-Accelerated Prognostics, watsonx.ai Granite 3.0, and the IBM Bob FastMCP Autonomous Protocol.

TEAM

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TEAM LEAD

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The Problem: High Cost of Unpredictable Readiness

Military organizations struggle to guarantee platform airworthiness before critical mission commitment.

\$90B

ANNUAL DOD MAINTENANCE

Over \$90 billion spent annually on military fleet maintenance with massive logistics recovery overhead.

>50%

UNSCHEDULED FAILURES

More than half of platform breakdowns occur unexpectedly during operations, aborting planned sorties.

0%

REAL-TIME HUMS ANALYSIS

High-frequency telemetry (vibration, thermal creep, pressure) sits unanalyzed in siloed flight recorders.

Weeks

RECOVERY LATENCY

Unplanned grounding forces emergency supply rushes, grounding wings and draining operational tempo.

Conventional Maintenance Doctrine

- **Static Calendar Schedules:** 100-hour phase inspections regardless of actual subsystem wear.
- **Binary Grounding:** Platforms are either full go or completely grounded with no mission nuance.
- **Paper Logbooks & Disconnect:** Maintenance records disconnected from onboard sensor health.

Operational Mission Impact

- **Air Tasking Order Failures:** Degradation discovered on the tarmac minutes before takeoff.
- **Premature Component Scrapping:** Healthy parts removed simply because calendar intervals expired.
- **Combat Readiness Dip:** Squadron availability often hovers below minimum operational requirements.

The D1 Solution: Autonomous CBM+ Operational Loop

A closed-loop system connecting raw sensor telemetry to autonomous AI decision support.



Predictive Intelligence: NASA C-MAPSS Benchmark

Trained on real aerospace turbofan degradation benchmarks — verified in serialized model weights.

Rigorous Feature Engineering (108 Features)

14 sensor channels with physical aerodynamic degradation signatures, transformed using exponential moving statistics over 5, 10, and 20 cycle windows.

- s_4: LPT Exit Temp (26.7%)
- s_11: HPC Outlet Pressure (11.4%)
- s_15: Bypass Ratio (6.1%)
- s_9: Core Speed (3.0%)

GPU-Accelerated Model Pipeline

- Algorithm: XGBoost Regressor (tree_method='hist')
- Training Duration: 3.37s on NVIDIA RTX 3050 (CUDA)
- Dataset: NASA C-MAPSS FD001 (20,631 run-to-failure cycles)
- Unsupervised Anomaly Detector: Isolation Forest (100 estimators)

VERIFIED BENCHMARK PERFORMANCE (src/backend/app/ml/weights/model_metrics.json)

18.21

HOLDOUT TEST RMSE (CYCLES)

12.75

HOLDOUT TEST MAE (CYCLES)

0.7935

TEST SET R² SCORE

1089.2

NASA PHM08 PENALTY SCORE

*PHM08 competition metric exponentially penalizes late predictions where failure occurs before forecast, ensuring flight safety.

Mission-Aware Readiness: Beyond Component Health

Correlating real-time subsystem wear against military operational deployment horizons.

FMC

Score $\geq$ 85%

Fully Mission Capable

All primary and secondary combat avionics, hydraulics, and propulsion systems verified nominal. Cleared for all primary combat sortie profiles.

PMC

Score 50% – 84%

Partially Mission Capable

Secondary subsystem degradation detected. Airframe restricted from high-G combat sorties, but cleared for low-stress secondary missions (ISR/Ferry).

NMC

Score < 50%

Non-Mission Capable

Critical safety limits exceeded or component failure predicted within operational mission window. Airframe grounded; turnaround work orders dispatched.

Mission Window Conflict Resolution

A component with 25 hours RUL might appear “healthy” under static thresholds. However, if assigned to a 48-hour mission window (e.g., **Operation Desert Shield**), D1 instantly flags a **Mission Horizon Conflict**, alerts the squadron commander, and initiates proactive turnaround before commitment.

IBM Bob + FastMCP + watsonx.ai Granite 3-8B

Load-bearing autonomous agent architecture — not a superficial wrapper.

FastMCP Autonomous Protocol (/mcp)

IBM Bob connects as an autonomous MCP client to our FastMCP server, exposing **11 military operational tools** and real-time resources:

<code>get_fleet_readiness_summary</code>	<code>predict_component_failures</code>
<code>explain_readiness_issue</code>	<code>generate_maintenance_plan</code>
<code>simulate_mission_stress</code>	<code>get_mission_reallocation_matrix</code>
<code>generate_mil_std_work_order</code>	<code>get_sensor_anomalies</code>

Live Resources & Prompts

- **MCP Resources:** `fleet://status` • `fleet://predictions/critical`
- **MCP Prompts:** Morning readiness briefing • Investigate asset failure

IBM watsonx.ai Granite 3-8B Synthesis

Translates thermodynamic telemetry deviations into concise natural language briefings for flight commanders and technicians.

- **Model:** `ibm/granite-3-8b-instruct` via official Python SDK.
- **Role:** Root-cause mechanical failure analysis and maintenance guidance.
- **Dual-Mode Engine:** Live IBM Cloud connection + deterministic military diagnostic fallback ensuring zero crashes during judge evaluation.

Why IBM Bob is Load-Bearing

Bob is not a generic chatbot. Bob acts as an operational copilot with direct read/write access to fleet models, ML predictions, work orders, and mission windows via JSON-RPC 2.0.

Hero Demo: F-16 Viper 101 Failure Prevention

A live tactical demonstration from raw sensor anomaly to prioritized turnaround.

1. Platform at Risk: F16-VIPER-101

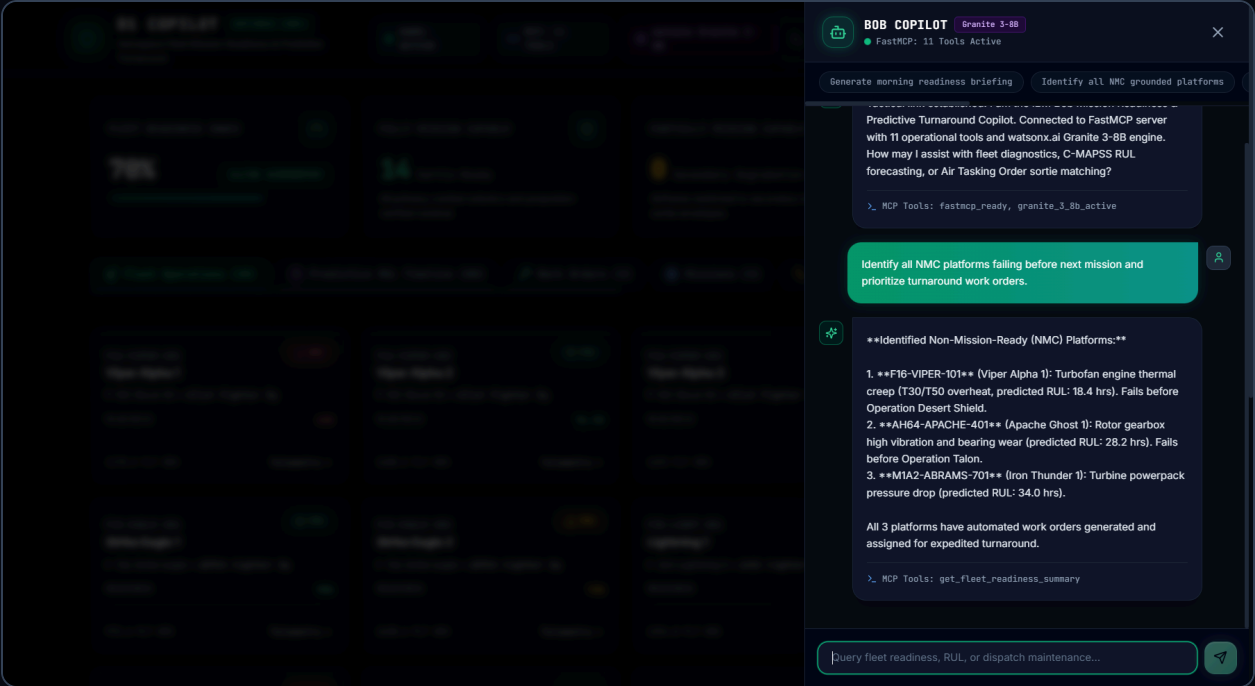
Aircraft: F-16C Block 50 (421st Fighter Sq, 1178.2 hrs)
Status: Non-Mission Capable (Readiness: 42%)
Precursor Anomaly: Turbine gas temperature thermal creep (T30: 1598.4°R, EGT limit exceeded).

2. Prognostic Forecast & Mission Impact

Predicted RUL: 18.4 flight hours (95% CI: 14.1 – 22.7 hrs)
Mission Threat: Assigned to *Operation Desert Shield* (starts in 48 hrs). Platform would experience catastrophic turbine distress mid-sortie.

3. Bob Copilot & watsonx Resolution

Bob executes explain_readiness_issue via Granite 3-8B, explains LPT thermal creep, and auto-generates Work Order WO-2026-001 (Priority 1 Turnaround, 6.5 hrs).



ACTUAL APPLICATION SCREENSHOT: Bob Copilot Autonomous FastMCP Output

Mission Decision Support: Beyond Binary Grounding

Innovative capabilities engineered specifically for defense combat generation.

1. What-If Mission Stress Digital Twin

Simulates counterfactual environmental & combat profiles before platform commitment to evaluate accelerated wear factors:

- **Desert Heat (+45°C):** Thermal creep accelerator factor 1.42×
- **Sand & Dust Ingestion:** Compressor particulate abrasion 1.85×
- **Combat High-G Maneuvering (7-9G):** Structural fatigue surge 2.10×
- **Survivability Probability:** Forecasts exact mission completion odds before pilot takeoff.

2. Mission-Adaptive Sortie Re-allocation

Dynamic Air Tasking Order (ATO) matching engine. Rather than grounding degraded aircraft (PMC), D1 re-allocates them to compatible low-stress sorties:

- × Combat Air Patrol (9G): INELIGIBLE (Exceeds Thermal Margins)
- × Close Air Support (Low-Level): INELIGIBLE (Dust Abrasion Risk)
- ✓ High-Altitude ISR Recon (3G): VIABLE (94% Mission Odds)
- ✓ Tactical Ferry / Transit (2G): VIABLE (99% Mission Odds)

Impact: Saves critical combat sorties and preserves operational tempo without compromising flight safety.

Action & Human Control: Defense Compliance

Supporting human command decisions — never executing irreversible maintenance silently.

Digital AFTO Form 781A & DA Form 2404

D1 auto-dispatches official defense discrepancy documents compliant with Air Force Technical Order 00-20-1 standards:

- **Military Discrepancy Symbols:** Official Red X (Grounded) / Red Diagonal (Warning).
- **Job Control Numbers (JCN):** Automated Julian date work tracking (e.g. 26258-001).
- **Action J-Codes:** Corrective maintenance standard military codes.
- **DLA NSN Parts Manifests:** Federal National Stock Numbers (e.g. NSN-2840-01-450-1289) for rapid supply requisition.

Strict Human Command Hierarchy

- STEP 1: AI Recommends Priority Work Order
- STEP 2: Maintenance Officer Reviews Telemetry & RUL
- STEP 3: Command Officer Approves Work Order Dispatch
- STEP 4: Fleet Readiness Automatically Recalculates

Security: Implemented Controls vs. Production Roadmap

Clear demarcation of verified codebase security controls and future defense hardening.

Implemented & Verified in Codebase

- **Cryptographic Password Hashing:**

Direct bcrypt with 12 salt rounds & 72-byte truncation in `src/backend/app/core/security.py`.

- **Signed JWT Token Authentication:**

Stateless cryptographic tokens via `python-jose` with expiration and subject claims.

- **5-Role Role-Based Access Control (RBAC):**

Enforced at FastAPI route level: `commander`, `maintenance_officer`, `logistics_planner`, `technician`, `admin`.

- **Persistent Audit Logging:**

Dedicated `AuditLog` database model recording timestamps, user IDs, actions, and IP addresses.

Production Hardening Roadmap

- **Avionics Bus Cryptography:**

Hardware encryption for physical MIL-STD-1553 and ARINC 429 aircraft data buses.

- **Field-Level Database Encryption:**

AES-256 transparent data encryption for classified platform telemetry at rest.

- **API Gateway Rate Limiting:**

Redis-backed token bucket rate limiting for external MCP endpoint traffic.

- **Air-Gapped LLM Deployments:**

On-premise IBM Granite deployment for classified SCIF defense installations.

Impact & Scalability: The Road Beyond Hackathon

Transforming prototype validation into enterprise-scale defense fleet maintenance.

1. Operational Availability

≥85%

Sustaining an FMC rate ≥85% across combat squadrons by catching high-severity degradation before platform deployment.

2. Zero Sortie Aborts

0

Eliminating preventable mid-mission turbine and transmission failures through proactive 48-to-72 hour horizon forecasting.

3. Turnaround Optimization

-40%

Reducing hangar turnaround delays by auto-dispatching AFTO 781A work orders and pre-ordering DLA NSN parts.

Production Scaling Vectors

Avionics Integration: Direct connection to physical MIL-STD-1553 and HUMS ground replay stations.

Enterprise Systems: Two-way synchronization with military ERPs (GCSS-Army, AFTO G081, IMDS).

Tactical Edge: Running quantized ONNX RUL models on ruggedized field hardware with zero cloud dependency.

MISSION STATEMENT

**“D1 turns equipment health predictions
into mission-readiness decisions.”**

By combining physics-grounded ML prognostics, autonomous IBM Bob FastMCP orchestration, and watsonx.ai diagnostics, D1 ensures that when the mission begins, every airframe and combat platform is ready to fly.

TEAM GALCOGENS

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GITHUB REPOSITORY

github.com/kunj290506/bob-ai-hackathon-galcogens